

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer specializing in **distributed systems**, **real-time data pipelines**, and zero-downtime cloud migrations. 5+ years building **scalable APIs** and **microservices** with Node.js and Python. Led end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime. Experienced with **Apache Kafka** for event streaming, **Docker** and **Kubernetes** orchestration, and building customer-facing analytics systems. Strong foundation in **cloud infrastructure** (GCP, Firebase, AWS) and systems programming (Rust).

CORE COMPETENCIES

- Distributed Systems
- Data Pipelines
- Apache Kafka
- Docker & Kubernetes
- Real-time Event Streaming
- Cloud Infrastructure (GCP)
- Zero-Downtime Migrations
- Microservices Architecture

WORK EXPERIENCE

MTailor September 2022 – Present

Software Engineer · US Remote

- Led end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, implementing **distributed systems** patterns for zero-downtime transition
- Built **real-time data pipeline** with two-way sync between MongoDB and Firestore using Node.js and **Google Pub/Sub** for message processing
- Implemented **observability** and monitoring for migration pipeline using GCP Cloud Monitoring, ensuring reliability throughout 6-month transition
- Built 3D visualization feature using video overlay and ffmpeg library, increasing buyer conversion
- Saved \$5,000/month by migrating all services off AWS to GCP, optimizing **cloud infrastructure** costs

MTailor March 2022 – July 2022

Contractor · US Remote

- Built migration script to read data from MongoDB and write to Firestore with complex processing logic for **distributed data** consistency
- Wrote documentation for engineering team on Firebase SDKs (Node.js and Python)
- Trained Ops team on Firebase Dashboard and backup procedures

Dr Wealth August 2021 – January 2022

Software Engineer · Singapore Remote

- Optimized **analytical queries** across 2M+ Firestore records for real-time stock price dashboards, improving **API performance**
- Built ad-hoc jobs to query large datasets, keeping customer prices current from Morningstar APIs
- Extended backend **APIs** using Firebase Cloud Functions and Express.js

Mind2matter.co January 2022 – July 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight timelines

CodeBits January 2020 – July 2021

Software Engineer · Uganda Remote

- Architected **microservices** backend for FIDA Uganda case management using **Apache Kafka** for event streaming, **Docker** containers, and **Kubernetes** orchestration
- Built USSD service for legal aid providers communicating with Node.js backend via gRPC
- Developed cross-platform mobile app using React Native and Expo with Firebase push notifications
- Led team of 4 developers building legal tech systems for NGOs

PROJECTS

Parse → Firebase Migration Platform Production

Zero-downtime migration system for 20+ applications. Built **real-time sync pipeline** with Pub/Sub, handling parallel production traffic during 6-month transition. Saved \$5K/month infrastructure costs.

Node.js, Firebase, GCP Pub/Sub, MongoDB, Python

Real-time Stock Analytics Dashboard Production

Optimized **analytical query** system across 2M+ records for real-time price updates. Integrated Morningstar APIs with caching strategy for performance.

Firebase, React, Express.js, Morningstar API

Legal Case Management Microservices Production

Event-driven **microservices architecture** for NGO case management. **Kafka** for messaging, **Kubernetes** for orchestration, gRPC for internal communication.

Node.js, Apache Kafka, Docker, Kubernetes, gRPC

EDUCATION

Makerere University — Bachelor of Science in Computer Engineering

2018 – 2022

Kampala, Uganda

SKILLS

Languages: Node.js, Python, JavaScript, SQL, Rust **Distributed Systems:** Apache Kafka, gRPC, Pub/Sub, Microservices

Cloud: GCP, Firebase, AWS (S3, EC2, EBS) **Containers:** Docker, Kubernetes **Databases:** MongoDB, Firestore, SQL

Frontend: React, React Native, Tailwind, HTML/CSS